Graphics Processing Units (Part 1)

Contents

Graphics Processing Unit

- August 31, 1999 - introduction of the Graphics Processing Unit (GPU) for the PC industry.
- Technical definition of a GPU: "a single chip processor with integrated transform, lighting, triangle setup/clipping, and rendering engines that is capable of processing a minimum of 10 million polygons per second." (NVIDIA site: http://www.nvidia.com/object/gpu.html).

GPU - Motivation

- In classic applications the CPU performs the most graphics operations (e.g. transformations for coordinate spaces, projections, clipping, rasterization, light computation, texture mapping etc).
- Necessity to offload graphics operations from CPU to specialized hardware for vector and matrix based processing.
- Application control over the stages in the graphics pipeline.
- GPU provides a general pipeline solution instead of specialized hardware for each graphics algorithm.
- Programmability and scalability.
- GPUs are used in:
  - embedded systems
  - mobile phones
  - personal computers
  - Workstations
  - game consoles

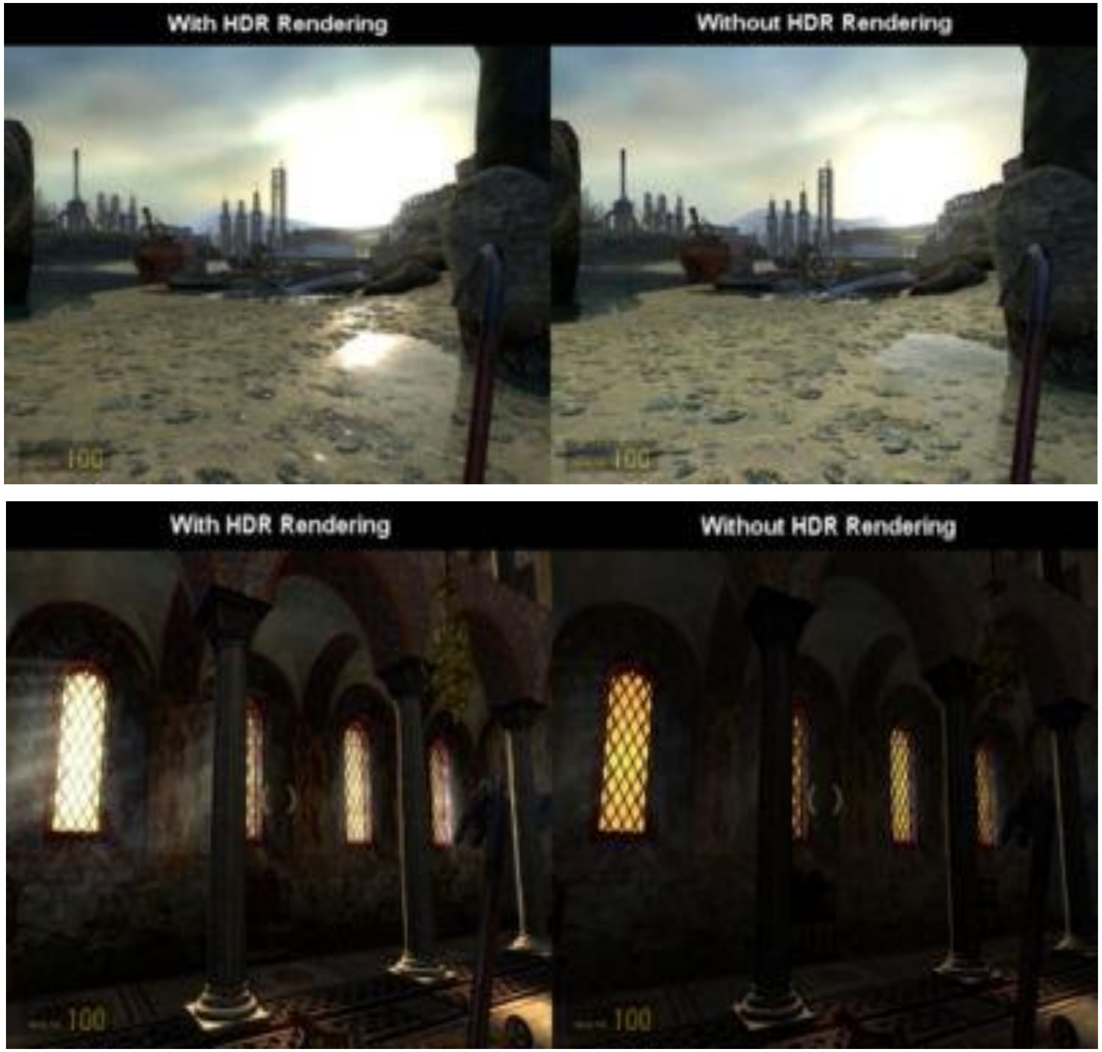
GPU based rendering

- High Dynamic Range Imaging (HDRI): Preservation of detail in large contrast differences

A comparison of HDR and LDR in the game *Far Cry*.

Wikipedia, "High Dynamic Range Rendering", Wikipedia , January 2008, <http://en.wikipedia.org/wiki/High_dynamic_range_rendering>

GPU based rendering



From *Half-Life 2: Lost Coast*, comparing HDR to LDR on how

(a) light is reflected.

(b) light passes through transparent materials.

Wikipedia, “High Dynamic Range Rendering”, Wikipedia , January 2008, <http://en.wikipedia.org/wiki/High_dynamic_range_rendering>

GPU based rendering

From *Serious Sam SE* , comparing Anisotropic Filtering (AF)

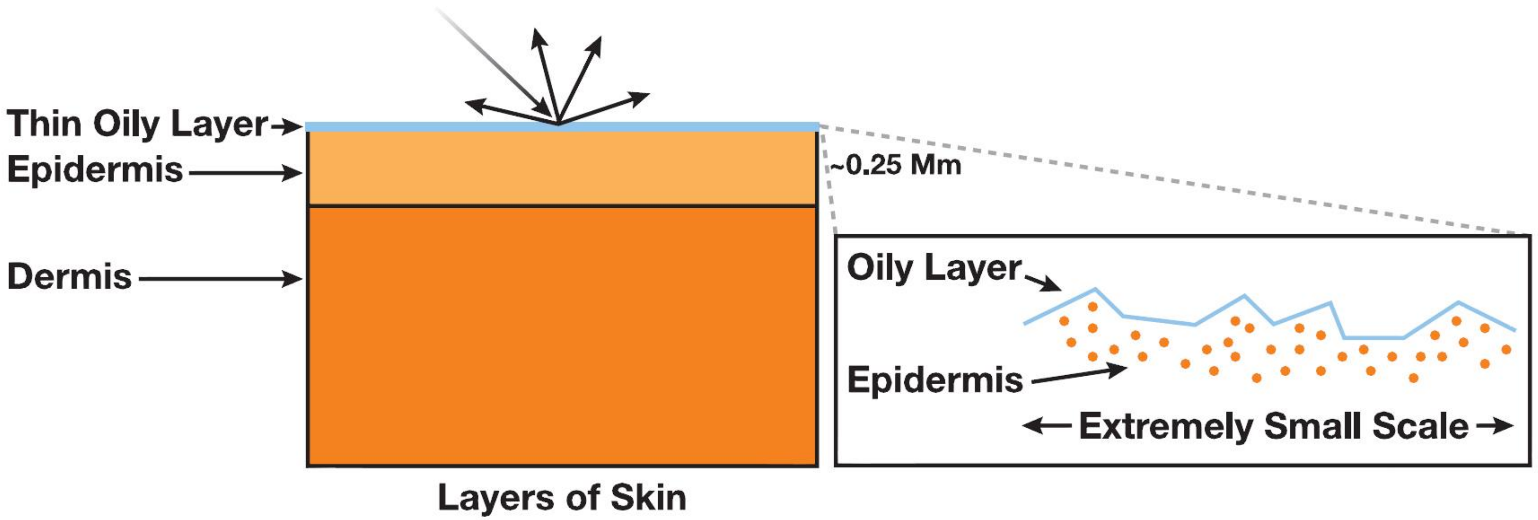
(a) with AF.

(b) without AF.

The Naked Truth About Anisotropic Filtering, January 2008, http://www.extremetech.com/article2/0,1697,1157434,00.asp

GPU based simulation and rendering

- The Appearance of Skin as materials of two components:
  1. Surface reflectance (specular)
  2. Subsurface reflectance (diffuse)
- Use Physically based models
  1. Specular BRDF
  2. Efficient subsurface scattering approximations

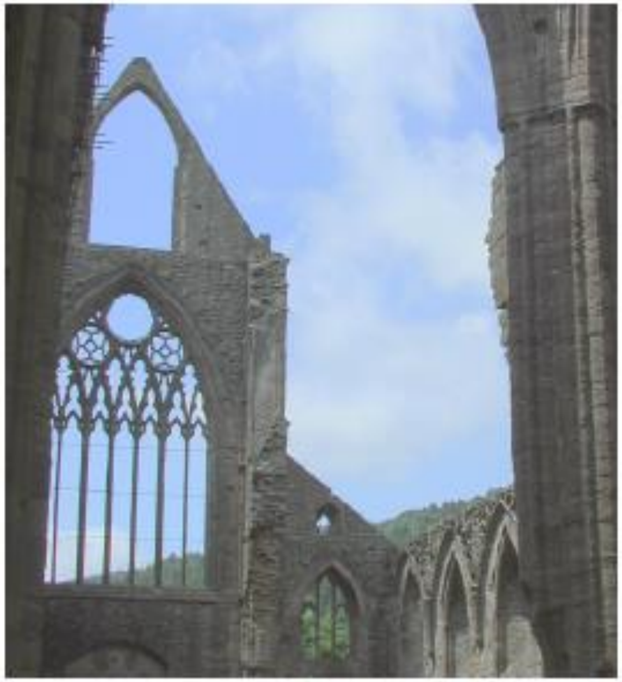


GPU based simulation and rendering

- Skin subsurface scattering
  - Gives skin soft appearance and color
  - Expensive to compute
  - Essential for realistic appearance

Eugene d'Eon, GPU Gems 3:Advanced Skin Rendering, SIGGRAPH 2007

GPU based rendering



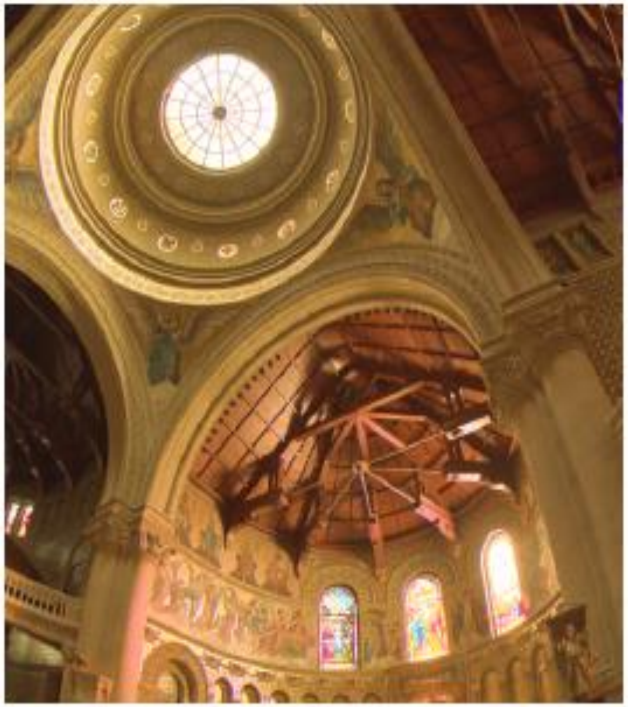
4 : 1

53 : 1

A series of 512x512 HDR images that have been tone mapped on the GPU. All images were generated at nearly 30 Hz.

Modern graphics architectures have replaced stages of the graphics pipeline with fully programmable modules. Therefore, it is now possible to perform fairly general computation on each vertex or fragment in a scene. In addition, the nature of the graphics pipeline makes substantial computational power available if the programs have a suitable structure.

GPU based rendering

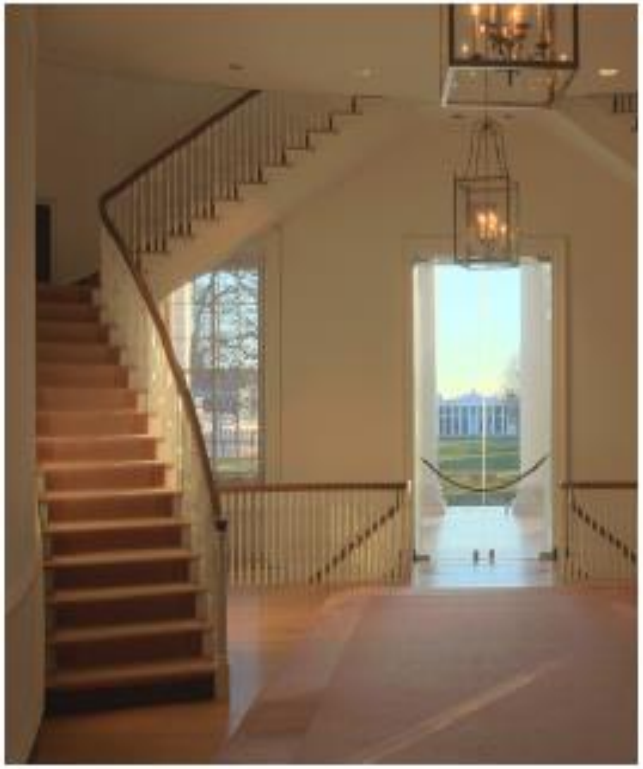


56 : 1

140 : 1

It allows an interactive application to achieve higher levels of realism by rendering with physically based, unclamped lighting values and high dynamic range texture maps. The tone mapping operator can be extended to include a time-dependent model, which is important for interactive behavior.

GPU based rendering



621 : 1

905 : 1

Ref: Nolan Goodnight, Rui Wang, Cliff Woolley, and Greg Humphreys, Interactive Time-Dependent Tone Mapping Using Programmable Graphics Hardware, Proceedings of the 14th Eurographics Workshop on Rendering, Per Christensen and Daniel Cohen-Or (Editors) 2003, pp.180-192.